

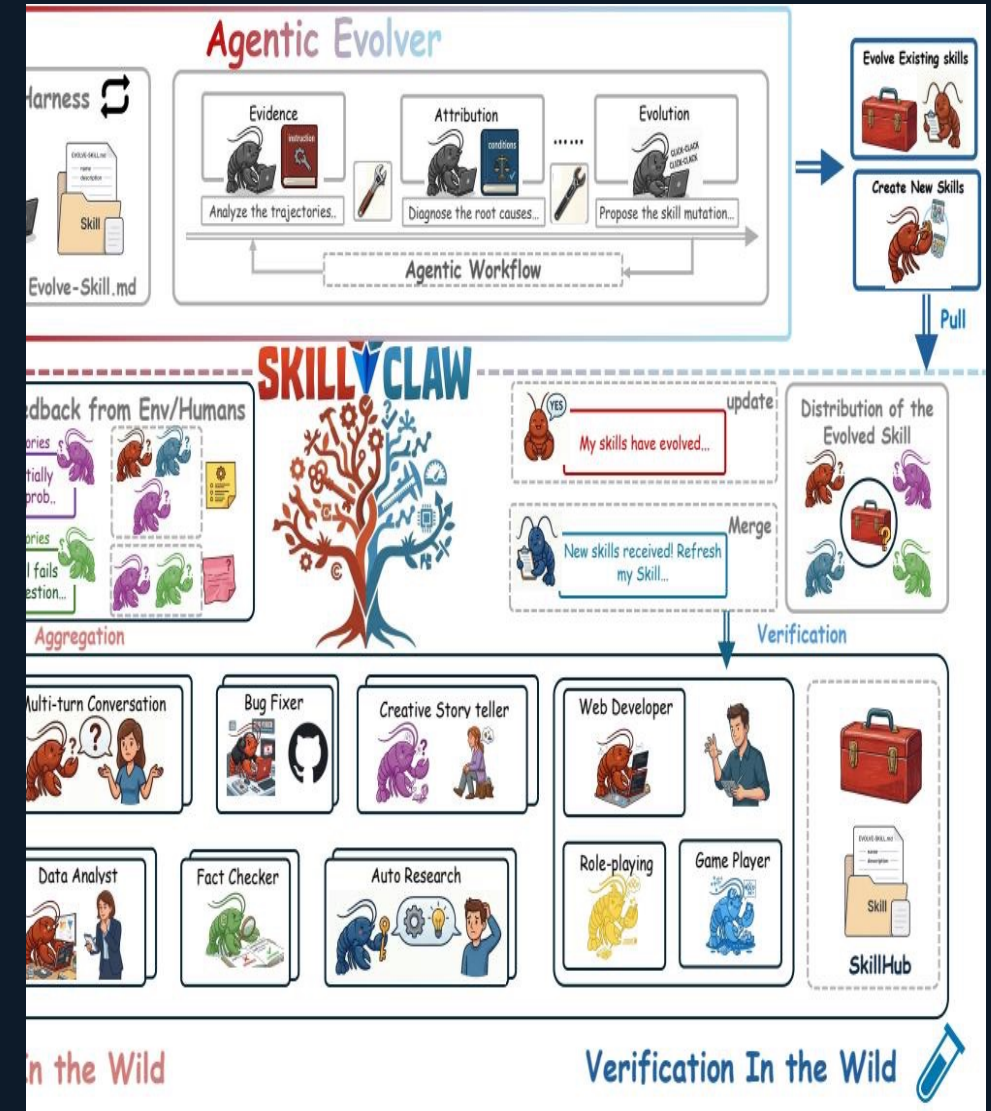
SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Let Skills Evolve Collectively with an Agentic Evolver

For ML Researchers & AI Systems Engineers

arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

STATIC SKILL PARADIGM

- Skills manually installed from a centralised hub
- No updates from runtime interaction
- Failure patterns re-encountered repeatedly across users
- Solutions never persist beyond individual sessions
- Every user rediscovers the same fixes independently
- Memory methods: tied to specific instances, hard to generalise
- Skill libraries: compressed but frozen — never evolve

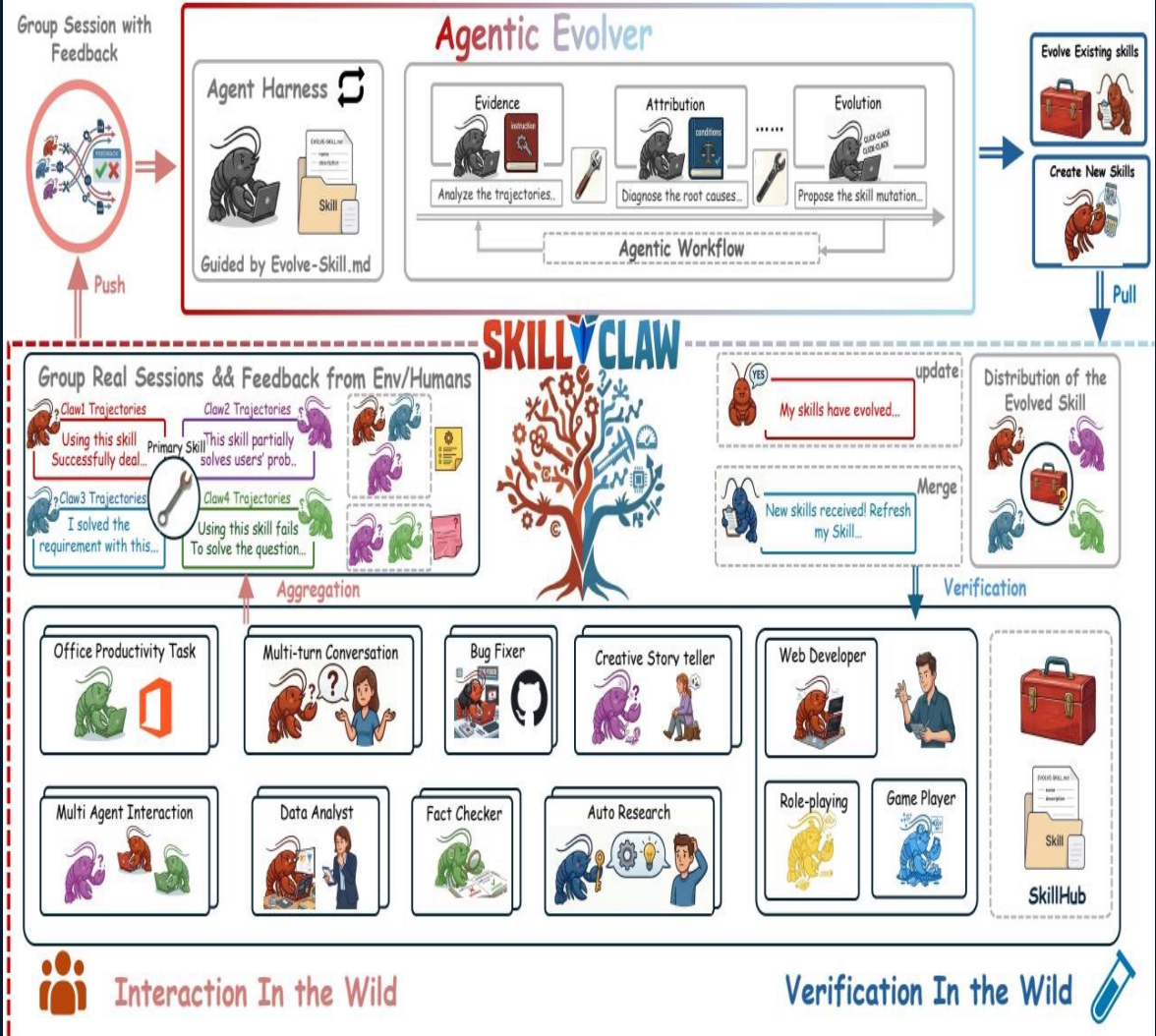
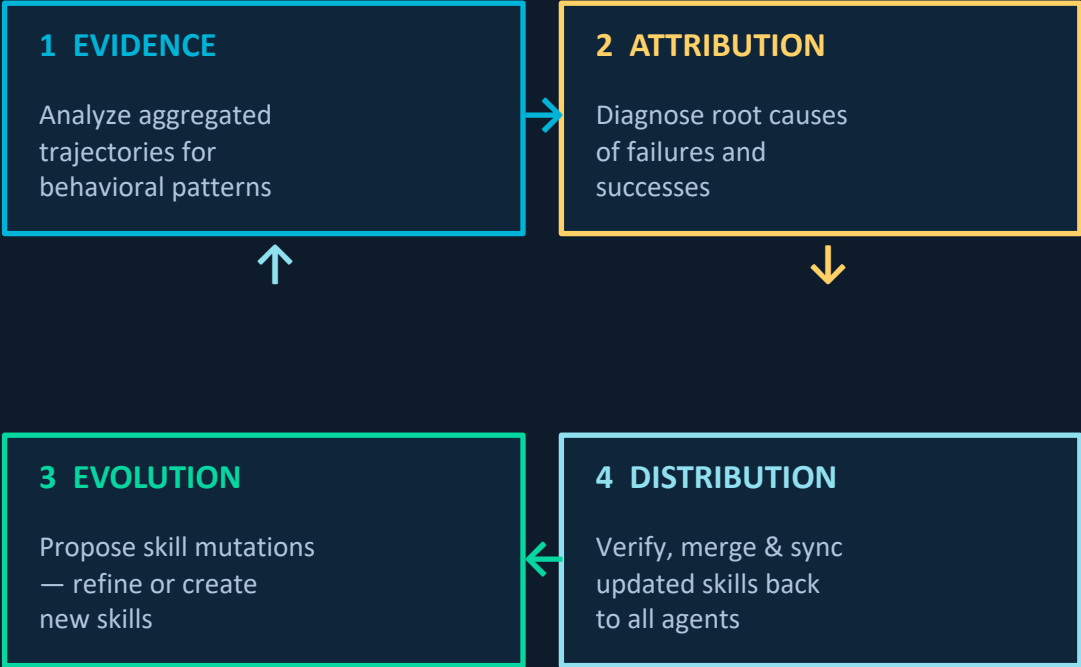


EVOLVING SKILL PARADIGM (SkillClaw)

- Skills evolve continuously from runtime interactions
- Cross-user trajectories feed a shared evidence base
- Failures trigger agentic diagnosis & skill mutation
- Improvements automatically synchronised to all agents
- Knowledge accumulates at the system level over time
- Open-ended reasoning — not predefined update rules
- Conservative verification prevents regressions

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution

SkillClaw Architecture Overview



Multi-user sessions → Session Collection → Agentic Evolver → Skill Synchronisation

Agentic Evolver: Evidence → Attribution → Evolution

Three-Stage Open-Ended Reasoning Pipeline — Not Predefined Rules

STAGE 1 EVIDENCE

- Aggregate session trajectories across all users
- Group by referenced skill
- Identify recurring behavioral patterns and edge cases
- Preserve full action–feedback causal chains



STAGE 2 ATTRIBUTION

- Diagnose root causes of observed failures
- Identify conditions under which skill succeeds
- Separate generalizable fixes from idiosyncratic patches
- Builds shared behavioral boundary model



STAGE 3 EVOLUTION

- Propose skill mutations via open-ended agent reasoning
- Refine existing skills OR create new ones
- Candidates verified before merge into repository
- Accepted skills sync to all connected agents

The evolver is an autonomous agent performing open-ended reasoning over interaction evidence — not a rule engine.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Complementary Signals From Diverse Contexts — Impossible From a Single User

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

Compatible Claw-style systems

Complementary Success Signals

Each user exercises the skill in a unique context. Successes under varied conditions reveal which aspects of the skill generalise broadly.

Diverse Failure Modes

Failures from different users expose distinct edge-cases that a single user's session would never surface — giving the evolver richer attribution data.

Generalizable vs. Idiosyncratic Fixes

Aggregation allows the evolver to distinguish a fix that helps everyone from one that only helps one user's workflow.

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Case Study: Multimodal Creative Pipeline — 6 Days of Evolution Attempts

Conservative verification: only 1 of 6 candidates accepted

Evolution Log — Multimodal Creative Task Pipeline Skill

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Only 1 of 6 candidates accepted — illustrates the conservative verification process that prevents skill regressions.

Case Study: Git Push Auth-Fallback — Iterative Refinement Over 6 Days

3 of 6 days yielded accepted updates — successful refinement before reaching a plateau

Evolution Log — git-push-with-auth-fallback Skill

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

3 of 6 days yielded accepted updates — iterative refinement succeeded before evidence plateaued and further proposals were rejected.

Key Takeaways: Three Properties That Distinguish SkillClaw

Implications for Continuously Improving Agent Systems

01

COLLECTIVE EVOLUTION

- Individual interactions contribute to a shared, continuously improving skill ecosystem
- Cross-user trajectories expose failure modes no single user session would surface
- Knowledge accumulates at the system level — not per-session

02

FULLY AUTOMATIC

- Skill evolution driven entirely by runtime interaction — zero manual curation
- No explicit user intervention required for skill updates to propagate
- Evaluation: WildClawBench + Qwen3-Max shows substantial cross-task improvement

03

AGENTIC EVOLUTION PARADIGM

- Updates via open-ended agent reasoning, not predefined rule templates
- Evidence → Attribution → Evolution pipeline enables context-aware mutations
- Compatible with OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw

SkillClaw demonstrates that LLM agent ecosystems can continuously self-improve — without human curation — through collective, agentic skill evolution.